

Enterprise Network Infrastructure Design

This repository contains a production-grade MikroTik RouterOS configuration, fiber distribution, and centralized services designed for a small-to-medium enterprise (SME) environment.

The configuration implements dual WAN connectivity, BGP routing, LAN segmentation, VPN services, NAT, and firewall-based traffic control.

Project Overview

This network design focuses on:

- High performance routing using MikroTik CCR series
- Fiber backbone for multi-floor distribution
- Segmented network architecture (CCTV, Users, Devices)
- Centralized services (NAS, NVR, IP PBX, VAT Server)
- PoE-based device deployment (Cameras, Access Points)

Network Architecture

- Logical separation using internal network
 - WAN bridge (public-facing interfaces)
- Multiple LAN subnets:
 - 192.168.10.0/24
 - 192.168.11.0/24
 - 192.168.20.0/22

Interface Organization

- ~~Physical interfaces~~ Dedicated ports for renamed for clarity and operational visibility
 - Access switches (SW1–SW4)
 - WAN uplinks
 - Core/internal connectivity

IP Addressing & DHCP

- ~~Defined IP ranges~~ Defined IP ranges for server with dynamic IP allocation
 - LAN clients
 - VPN users
- Google Public DNS configured (8.8.8.8, 1.1.1.1)

Core Components

◆ Routing Layer

- MikroTik CCR2004 Router
- Dual WAN capability (Primary & Secondary)
- SFP uplink to core switch

◆ Core Switching

- MikroTik CRS328 Switch
- Fiber uplinks to floor switches
- Central distribution point

◆ Servers & Services

- VAT Server
- Synology NAS (Storage & Backup)
- 64 Channel NVR (CCTV Recording)
- UCM6202 (IP PBX System)

Network Segmentation

Network Type	Subnet	Purpose
Static Network 1	192.168.10.0/24	Printers, IP Phones, Access Ctrl
Static Network 2	192.168.11.0/24	CCTV Cameras
DHCP Network	192.168.20.0/22	WiFi & LAN Users

Cabling Design

Cable Type	Purpose
Blue CAT6	ISP → Router / Router → Devices
Ash CAT6	Router → Servers
Orange CAT6	PoE Devices (Cameras, APs)
White CAT6	General Device Connections
Optical Fiber	Floor-to-floor backbone

Floor Distribution

Each floor contains:

- 48-Port PoE Switch (CCTV + APs)
- 24/48-Port Non-PoE Switch (Users)
- Fiber uplink to core switch
- Structured separation between:
 - Cameras

- User devices
- Network equipment

Key Features

- ✓ Stable LAN-based design
- ✓ Centralized monitoring & storage
- ✓ Efficient PoE deployment
- ✓ Fiber backbone for high bandwidth
- ✓ Suitable for enterprise & campus networks

Use Cases

- Corporate office network
- Surveillance-heavy environments
- Smart building infrastructure
- Campus networking

Contribution

Feel free to fork, modify, or use this architecture for:

- Learning purposes
- Real deployments
- Network planning
- Documentation reference

VPN Services

- RADIUS and 128-bit AES VPN support
- Dedicated VPN IP pool for client isolation

Firewall & Traffic Control

- Layer 7 protocol-based media filtering
 - Blocks social media (Facebook, YouTube)
 - Blocks financial/stock-related websites
 - Address-list based traffic control
 - Basic LAN ↔ VPN communication policies
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NAT Configuration

- Source NAT using public IP (103.190.199.153)
 - NAT bypass for internal/private routing (No-NAT list)
 - Masquerade for VPN clients
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Dynamic Routing (BGP)

- BGP configuration applied with two upstream ISPs
 - Primary link preferred (local-pref 1000)
 - Secondary link as backup (local-pref 100)
 - Controlled advertisement of public prefix:
 - 103.190.199.152/29
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Security Hardening

- Disabled unnecessary services:
 - Telnet, HTTP, SSH, API
 - Restricted Winbox access (custom port 9900)
 - Limited access to trusted IP ranges only
 - Logging buffer increased for better monitoring
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System & Monitoring

- NTP synchronization enabled
 - System identity configured (Reverie HO)
 - Logging and housekeeping IP references included
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Use Case

This configuration is suitable for:

-  Corporate headquarters network
 -  ISP edge router
 -  Industrial/enterprise environments
 -  Remote-access enabled infrastructure
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Notes & Recommendations

- Even if the degree of confidentiality is not as high as for production security
- Always secure BGP sessions with authentication (MD5) in production
- Regular backups and RouterOS updates are strongly advised

Network Overview

- Dual ISP uplinks with BGP failover
- Central MikroTik router handling routing, NAT, firewall, and VPN
- LAN distribution via multiple switches
- Remote users connected via VPN

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